

Label Cover to CVP reduction

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1 Reformulation of CVP

Definition 1.1. (*Problem X*)

Let $\mathbf{y} = (y_1, y_2, \dots, y_n) \in \mathbb{Z}^n$. Consider a non-homogeneous system:

$$\psi_k(\mathbf{y}) = c_k + \sum_j^n b_{kj}y_j, \quad 1 \leq k \leq t$$

Consider a homogeneous system:

$$\phi_i(\mathbf{y}) = \sum_j^n a_{ij}y_j, \quad 1 \leq i \leq m$$

The problem is to optimize the following:

$$\min_{\mathbf{y} \in \mathbb{Z}^n} \left(\sum_{i=1}^m |\phi_i(\mathbf{y})|^2 \right)^{1/2}$$

subject to $\psi_k(\mathbf{y}) = 0 \quad \forall 1 \leq k \leq t$

Remark 1. We don't need $\phi_i(\cdot)$ per se to define Problem X. But we are using this because we might need it in our reduction from some other problem to Problem X.

1.1 Reduction from Problem X to CVP

Let us assume Problem X has a particular solution $\mathbf{y}^0$ with integer coordinates. That is, we are saying the non-homogeneous system $\psi_k(\mathbf{y}) = 0$ has atleast one integral solution. Consider the homogeneous system corresponding to $\psi_k(\cdot)$:

$$\begin{aligned} \theta_k(\mathbf{y}) &= \psi_k(\mathbf{y}) - c_k \\ \Rightarrow \theta_k(\mathbf{y}) &= \sum_j b_{kj}y_j, \\ &\quad \forall 1 \leq k \leq t \end{aligned}$$

$$\begin{aligned} \psi_k(\mathbf{y}^0) = 0 &\Rightarrow c_k + \sum_j b_{kj}\mathbf{y}_j^0 = 0 \\ &\Rightarrow c_k = - \sum_j b_{kj}\mathbf{y}_j^0 \end{aligned} \tag{1}$$

Let $\mathcal{B}$ be the lattice corresponding to integral solutions of $\theta_k(\cdot) = 0$. Further let $\mathbf{y}^1$ be any arbitrary solution to the system $\psi_k(\cdot) = 0$.

$$\begin{aligned}
& \Rightarrow \psi_k(\mathbf{y}^1) = 0 \\
& \Rightarrow c_k + \sum_j b_{kj} \mathbf{y}_j^1 = 0 \\
& \Rightarrow - \sum_j b_{kj} \mathbf{y}_j^0 + \sum_j b_{kj} \mathbf{y}_j^1 = 0 \text{ [Using (1)]} \\
& \Rightarrow \sum_j b_{kj} (\mathbf{y}_j^1 - \mathbf{y}_j^0) = 0 \\
& \Rightarrow \theta_k(\mathbf{y}^1 - \mathbf{y}^0) = 0 \\
& \Rightarrow \mathbf{y}^1 - \mathbf{y}^0 \in \text{Lattice } \mathcal{B}
\end{aligned} \tag{2}$$

Thus, any solution of the system $\psi_k(\mathbf{y}) = 0$ is of the form $\mathbf{y}^0 - l$ where l is an element of the Lattice $\mathcal{B}$. (For solution $\mathbf{y}'$ of $\psi_k(\cdot) = 0$, $l = \mathbf{y}^0 - \mathbf{y}'$)

Problem X corresponds to minimizing $\|\phi \mathbf{y}\|$ over $\mathbf{y} \in \mathbb{Z}^n$ subject to $\psi \mathbf{y} = 0$. This is same as minimizing $\|\phi(\mathbf{y}^0 - l)\|$ over $l \in \mathcal{B}$. This is same as minimizing the distance of point $\phi \mathbf{y}^0$ from the Lattice $\phi \mathcal{B}$. Hence, we get an instance of CVP

Remark 2. $\mathbf{y}^0$ is, in general, not a point on Lattice $\mathcal{B}$.

Remark 3. We don't care about the case when the system $\psi_k(\cdot) = 0$ has no integral solutions because even Problem X is vacuous in that case.

2 Label Cover to Problem X reduction

Definition 2.1. (*Label Cover*)

$G = (V, W, E)$ is a bipartite graph. V : left vertices, W : right vertices, E : edges. The graph is left regular i.e., all vertices in V have the same degree. $n = |V|$ and $m = |W|$.

We have to assign one **label** to every vertex. Vertices in V receive label from set $[R]$ and vertices from W receive label from set $[S]$. A labelling A is a map. $A : V \mapsto [R], W \mapsto [S]$.

The labelling is supposed to satisfy constraints given by maps $\pi_{v,w} : [R] \mapsto [S]$. There is one such map for every edge $(v, w) \in E$

Remark 4. There are multiple maps, one for each pair of vertices (v, w) . It is like if (v, w) are supposed to have labels from $\{(r_1, s_1), (r_2, s_2), \dots, (r_\mu, s_\mu)\}$.

A labelling A **satisfies** an edge (v, w) , if

$$\pi_{v,w}(A(v)) = A(w)$$

Remark 5. We require the maps $\pi_{v,w}$ to be functions (or single-valued maps);

Remark 6. In this setting, a labelling must assign each vertex **exactly one** label. Nothing is said about whether 2 vertices can have same label. It is allowed most likely.

The optimum $OPT(LC)$ of the instance is defined to be the maximum fraction of edges satisfied by any labelling.

We assume $n \geq m$, and $R \geq S$. (Left side is larger)

Definition 2.2. (GapCVP)

γ – GapCVP is a promise problem $(\mathcal{L}(\mathbf{B}), \mathbf{t}, r)$ whose YES instances satisfy $\text{dist}(\mathbf{t}, \mathcal{L}(\mathbf{B})) \leq r$, and NO instances satisfy $\text{dist}(\mathbf{t}, \mathcal{L}(\mathbf{B})) \geq \gamma r$.

Theorem 2.3. [*Kho09*] There exists an absolute constant $\beta > 0$ such that for every integer $R \geq 7$, there is a reduction from SAT instance ϕ to Label Cover instance $LC(G(V, W, E), n, m, \{\pi_{v,w}\}, [R], [S])$ with the following property:

- The YES instances of SAT map to label cover instances with $OPT(LC) = 1$.
- The NO instances map to label cover instances with $OPT(LC) \leq R^{-\beta}$.

The running time of the reduction and the size of the label cover instance are bounded by $|\phi|^{O(\log R)}$.

Remark 7. Absolute constant means β is independent of R .

Remark 8. The only freedom we have is in choosing R . Note that the large integer R is same used in left-label Universe $[R]$.

2.1 Reduction

Let $LC(G(V, W, E), n, m, \{\pi_{v,w}\}, [R], [S])$ be the instance of Label Cover given by *Thm 2.3*(2.3).

Remark 9. This is in essence an instance of GapLC.

We describe a reduction from this instance to an instance of $\gamma - \text{GapCVP}$ with gap $\gamma = \frac{1}{20}R^{\beta/2}$.
Remark 10. Note that we have freedom with R . So we can choose appropriate gap by adjusting R . Instead of giving a direct reduction to GapCVP, we give a reduction to *Gap* variant of Problem X.

2.1.1 Problem X construction

The set of integer valued variables ($\mathbf{y}$ in Problem X) is:

$$Y := \{y_{v,j} \mid v \in V, j \in [R]\} \cup \{z_{w,i} \mid w \in W, i \in [S]\}$$

The function to be minimized (optimization problem in Problem X) is:

$$OBJ := \left(m \cdot \sum_{v \in V, j \in [R]} y_{v,j}^2 + n \cdot \sum_{w \in W, i \in [S]} z_{w,i}^2 \right)^{1/2}$$

The *affine* constraints ($\psi_k(\cdot) = 0$ in Problem X) are:

$$\forall v \in V, \sum_{j \in [R]} y_{v,j} = 1 \quad (3)$$

$$\forall w \in W, \sum_{i \in [S]} z_{w,i} = 1 \quad (4)$$

$$\forall (v, w) \in E, \forall i \in [S], z_{w,i} = \sum_{j \in [R]: \pi_{v,w}(j)=i} y_{v,j} \quad (5)$$

2.1.2 Remarks

Remark 11. There are $n \cdot R + m \cdot S$ variables.

Remark 12. These variables can take all integer values, not just 0 and 1.

Remark 13. These variables don't represent anything, in general. In particular, these variables are not to be seen as Indicator variables.

Remark 14. In case of a YES instance of GapLC, there exists a 0,1 assignment of these variables which would then correspond to an optimum labeling ($OPT(LC) = 1$ labelling). The same assignment corresponds to optimum of Problem X as well. ($OBJ = \sqrt{2mn}$)

Remark 15. In case of NO instance of GapLC, we argue that every integer assignment of these variables results in OBJ value much greater than $\sqrt{2mn}$.

Remark 16. To figure out: Why m and n weights attached to summations in OBJ . Maybe to get an expression like $\sqrt{2mn}$ that we can easily handle. But are m and n necessary for the reduction?

Remark 17. If we consider only 0,1 assignments, we get a 1-1 correspondence of assignments with valid labellings.

Remark 18. If we consider only 0,1 assignments to variables, then YES instance will have a 0,1 assignment that satisfies all constraints (3)-(5). This is because YES instance of GapLC has $OPT(LC) = 1$.

Remark 19. In case of NO instance of GapLC, no labelling can satisfy all the edges. Thus, no 0,1 assignment can satisfy all constraints (3)-(5).

Remark 20. We have 2 options of dealing with the issue of NO instances:

- Allow some constraints amongst (3)-(5) to be unsatisfied.
- Expand the solution space to allow the variables to take integer values other than 0,1 as well. But the variables must satisfy all the constraints (3)-(5).

First option doesn't give us good traction on the problem. There are too many degrees of freedom. What constraints are non-negotiable, what are negotiable? Do they capture the essence of Label Cover problem? These questions don't give positive answers.

One thing we can try is: “What is OBJ for an assignment that satisfies maximum number of constraints (3)-(5).” This leads to a dead-end. What constraints are negotiable? If (3),(4) are negotiable, then the assignment no longer represents a labelling with exactly one label to each vertex. On the other hand, if (3),(4) are non-negotiable, then OBJ is always equal to $\sqrt{2mn}$. This destroys the *gap* between YES/NO instances.

Hence, we consider the second option. We allow the variables to take integer values other than 0,1 as well. This relaxation would give assignments that might not represent a Label Cover instance on face value. But we prove that this relaxation preserves the YES/NO instances. YES instances still have $OBJ \leq \sqrt{2mn}$ and NO instances still have $OBJ \geq 0.1 \cdot R^{\beta/2} \sqrt{mn}$. (Recall that we have freedom with R . Thus, we can choose R big enough to get desired gap.)

Remark 21. We can have the case that no integer assignment to the variables satisfies all constraints (3)-(5). This is a vacuous case for which we can adequately define $OBJ := +\infty$.

Remark 22. To figure out: The reduction in reverse direction from this particular variant of Problem X to GapLC seems unlikely (atleast direct reduction seems infeasible). A particular instance of Problem X might have optimum OBJ corresponding to some *non-binary* integer assignments to variables. Such assignments don't have a direct correspondence in the Label Cover Problem in present setting.

Remark 23. Let us go back to first principles. A reduction from problem A to problem B means that if we have a subroutine to solve Problem B, then we can use it to solve Problem A. In a sense an arbitrary instance of Problem A can be converted to an instance of Problem B, such that the YES instance of Problem A map to YES instance of Problem B, and NO instances of Problem A map to NO instances of Problem B.

Now Problem A is Gap-Label Cover, where gap means YES instances have a labelling that covers all edges and NO instances cannot have more than $R^{-\beta}$ fraction of edges covered for any labelling. Here, the YES/NO instances are preserved, and the gap is transformed from $R^{-\beta}$ to $\frac{1}{20}R^{\beta/2}$. β is an absolute constant and $R \geq 7$ is an integer that can be chosen as large as desired.

2.1.3 YES Case

We prove that if the Label Cover has a labelling that satisfies all edges, i.e., $OPT(LC) = 1$, then there is an integer assignment to the variables in Y such that $OBJ \leq \sqrt{2mn}$.

Let $A : V \mapsto [R], A : W \mapsto [S]$ be such a labelling. Define:

$$y_{v,j} = \begin{cases} 1 & \text{if } j = A(v) \\ 0 & \text{otherwise} \end{cases}$$

$$z_{w,i} = \begin{cases} 1 & \text{if } i = A(w) \\ 0 & \text{otherwise} \end{cases}$$

Since there is exactly one label for each vertex, for each $v \in V$, exactly one $y_{v,j} = 1$ and rest all are 0. Similarly, for each $w \in W$, exactly one $z_{w,i} = 1$ and rest all are 0. Therefore,

$$OBJ = \sqrt{m \cdot |V| + n \cdot |W|} = \sqrt{2mn}$$

Exactly one label condition for each vertex ensures (3) and (4) are satisfied. Now, we show that all constraints in (5) are satisfied. Consider any $(v, w) \in E$ and $i \in [S]$. We will show that

$$z_{w,i} = \sum_{j \in [R] : \pi_{v,w}(j) = i} y_{v,j} \quad (6)$$

Let $i_0 = A(w)$ and $j_0 = A(v)$. This implies $z_{w,i_0} = 1$, $y_{v,j_0} = 1$. Since the labelling satisfies all edges, it satisfies edge $(v, w) \in E$ in particular as well. Thus, we have $\pi_{v,w}(j_0) = i_0$. Moreover, since each vertex has exactly one label, we have $\forall i \neq i_0, z_{w,i} = 0$ and $\forall j \neq j_0, y_{v,j} = 0$. Hence, (6) holds for $i = i_0$.

For $i \neq i_0$, LHS = 0. Since $\pi_{v,w}$ is a function, therefore $\pi_{v,w}(j_0)$ gives atmost 1 output. (exactly one here, $\pi_{v,w}(j_0) = i_0$) Thus, for $i \neq i_0$, the summation does not include y_{v,j_0} . Since $\forall j \neq j_0, y_{v,j} = 0$, we have RHS = 0 as well. Hence, (6) holds for $i \neq i_0$ too.

Thus, in YES case, we explicitly give an assignment satisfying (3)-(5).

2.1.4 NO case

Let $\alpha = R^{-\beta}$. Since we have freedom with the choice of R , we can choose R large enough to ensure $\alpha < 0.1$. Hereafter we assume $\alpha < 0.1$.

We prove that if the Label Cover instance has no labelling that satisfies even α fraction of its edges, then for any integer assignment to variables in Y (*any means positive, negative, non-binary all allowed*) that satisfies *all* constraints (3)-(5), one must have:

$$OBJ \geq 0.1 \sqrt{mn/\alpha}$$

Since we are considering Label Cover instances given by (2.3), NO instances will have $OPT(LC) \leq R^{-\beta}$. Thus, if the aforementioned result is proven, we will have:

$$OBJ \geq 0.1 \cdot R^{\beta/2} \sqrt{mn}$$

Therefore the gap between YES and NO cases is:

$$\frac{0.1 \cdot R^{\beta/2} \sqrt{mn}}{\sqrt{2mn}} \geq \frac{1}{20} R^{\beta/2} \quad \text{as claimed}$$

Proof.

Consider any *integer* assignment A to variables in Y that satisfies *all* constraints (3)-(5). Define sets $T_v \subseteq [R]$, $T_w \subseteq [S]$ as:

$$T_v := \{j \in [R] \mid y_{v,j} \neq 0\}, \quad T_w := \{i \in [S] \mid z_{w,i} \neq 0\}$$

Due to constraints (3) and (4), the sets T_v, T_w are non-empty for all $v \in V$, $w \in W$.

Remark 24. In essence, the sets T_v, T_w represent what variables got assigned non-zero values corresponding to each vertex.

Remark 25. Note that these variables don't correspond to any labelling of vertices per se. We will later show that any labelling can be visualised in this setting. This assignment problem, infact, is a superset of all possible labellings. We are proving that even for this superset, we always get $OBJ \geq 0.1\sqrt{mn/\alpha}$. Then it implies, for all valid labellings as well, we get $OBJ \geq 0.1\sqrt{mn/\alpha}$.

Remark 26. If no integer assignment to the variables in Y satisfies all constraints (3)-(5), then $OBJ := +\infty$. This correctly corresponds to **NO** instance of Label Cover.

Lemma 2.4. *For any $(v, w) \in E$ and $i \in T_w$, there exists $j^* \in T_v$ such that $\pi_{v,w}(j^*) = i$.*

Proof.

Consider the constraint (5) $z_{w,i} = \sum_{j \in [R]: \pi_{v,w}(j)=i} y_{v,j}$. Since $i \in T_w$, we have $z_{w,i} \neq 0$. Now, $z_{w,i} \neq 0$, in turn, implies atleast one of the variables on the right side of the constraint (5) must be non-zero. Let y_{v,j^*} be such a non-zero variable. Since $y_{v,j^*} \neq 0$, we have $j^* \in T_v$. Thus, we have $j^* \in T_v$ and $\pi_{v,w}(j^*) = i$, as desired.

Remark 27. The Lemma (2.4) in essence says, for each edge, if the right endpoint gets some non-zero assignment to a variable, then there has to be a corresponding variable under $\pi_{v,w}$ getting non-zero assignment on the left endpoint as well. In other words, for any edge, right endpoint gets no non-zero assignment without a corresponding non-zero assignment on left endpoint. This is basically a reformulation of Constraint (5) in new setting.

We consider 2 cases depending on whether the typical size of sets T_v is *small* or *large*. Let

$$V^* = \{v \in V \mid |T_v| \geq 0.1/\alpha\}.$$

Remark 28. Since we assume $\alpha < 0.1$, we have $0.1/\alpha > 1$. Thus, the set V^* is non-trivial. (Recall that each set T_v is non-empty, so $|T_v| \geq 1$ would have been trivially true for each v .)

Case 1 $|V^*| \geq 0.1|V| = 0.1 \cdot n$. We have

$$\begin{aligned}
OBJ &= \left(m \cdot \sum_{v \in V, j \in [R]} y_{v,j}^2 + n \cdot \sum_{w \in W, i \in [S]} z_{w,i}^2 \right)^{1/2} \\
&\geq \left(m \cdot \sum_{v \in V, j \in [R]} y_{v,j}^2 \right)^{1/2} \\
&\geq \left(m \cdot \sum_{v \in V^*, j : y_{v,j} \neq 0} 1 \right)^{1/2} \\
&= \left(m \cdot \sum_{v \in V^*} |T_v| \right)^{1/2} \\
&\geq \sqrt{m \cdot |V^*| \cdot 0.1 / \alpha} \\
&\geq 0.1 \sqrt{mn / \alpha}
\end{aligned} \tag{7}$$

Thus, we get $OBJ \geq 0.1 \sqrt{mn / \alpha}$, as desired.

Case 2 $|V^*| < 0.1|V| = 0.1 \cdot n$.

Ignore the edges (v, w) that are incident on V^* . (V^* is a set consisting of left vertices. We are ignoring the edges that have left endpoint as a vertex from V^*) Since, the graph is left regular, each left vertex has same degree. Therefore, by ignoring aforementioned edges (corresponding to $< 0.1|V|$ left vertices), we are ignoring atmost 0.1 fraction of the edges.

Now we give a labelling that covers much more than α fraction of the edges, thereby arriving at a contradiction (Since, in a NO instance, no labelling covers more than α fraction of edges)

Define the following labelling to V and W . The label of $w \in W$ is an arbitrary label from set T_w . The label of $v \in V$ is a random label from set T_v . We show that this labelling satisfies, on expectation, at least $0.9\alpha/0.1$ edges. Consider any arbitrary edge (v, w) such that $v \notin V^*$. Since $v \notin V^*$, we have $|T_v| \leq 0.1/\alpha$. Let label of w be some $i^* \in T_w$. By *Lemma (2.4)*, we know that there exists a label $j^* \in T_v$ such that $\pi_{v,w}(j^*) = i^*$. With probability $1/|T_v| > \alpha/0.1$, we select j^* as the label of v and the edge (v, w) is satisfied. (Since we are *randomly* selecting label of v from T_v , the probability of landing on right label is $1/|T_v|$. Further since $|T_v| < 0.1/\alpha$ for $v \notin V^*$, we get $1/|T_v| > \alpha/0.1$)

Thus, each edge is satisfied with probability $> \alpha/0.1$. There are $> 0.9|E|$ edges with left vertex not in V^* . Since, the label assignment corresponding to any edge doesn't care about other edges, the assignments are independent. Therefore, on expectation $> \frac{0.9\alpha}{0.1}E$ edges are satisfied, i.e., $> 9\alpha$ fraction of edges are satisfied, a contradiction. Therefore Case 2 cannot occur. This completes the proof.

Remark 29. In YES case, we need to show “there exists” an assignment that gives $OBJ \leq \sqrt{2mn}$. That is done by explicitly giving the assignment.

Remark 30. In NO case, we show that “for all” assignments to variables in Y , $OBJ \geq 0.1\sqrt{mn/\alpha}$.

2.1.5 Technical Analysis

Let n be the size of SAT instance. Combining *Thm(2.3)* with the Reduction from Label Cover to GapCVP (*via Problem X*), we get a reduction from SAT to GapCVP with $Gap = \frac{1}{20}R^{\beta/2}$ that runs in time $n^C \log R$ for some constant C .

Since we have freedom with choice of R , we choose $R = 2^{(\log n)^k}$ for some large enough integer k . (Note that we are shifting the freedom of choice from R to k) The size of CVP instance is obviously less than the time allowed for reduction. Let N denote the size of CVP instance. We have $N \leq n^C \log R$. Thus $\log N \leq C \log R \log n$. Substituting $R = 2^{(\log n)^k}$, we get:

$$\begin{aligned}
\log N &\leq C \cdot (\log n)^k \cdot \log n \\
&\leq C \cdot (\log n)^{k+1} \\
&\leq (\log n)^{k+2} \quad [\text{Since constant } C \leq \log n.] \\
&\quad \underline{\text{Constraint not tight}} \\
&\Rightarrow (\log N)^{1/(k+2)} \leq \log n
\end{aligned} \tag{8}$$

Now, consider the gap:

$$\begin{aligned}
Gap &= \frac{1}{20}R^{\beta/2} = \frac{1}{20}2^{(\beta/2) \cdot \log R} \\
&= \frac{1}{20}2^{(\beta/2) \cdot (\log n)^k} \\
&\geq \frac{1}{20}2^{(\log n)^{k-1}} \quad [\text{Since } \beta/2 \geq \log n] \quad \underline{\text{Constraint not tight}} \\
&\geq 2^{(\log N)^{(k-1)/(k+2)}}
\end{aligned} \tag{9}$$

Since we are choosing k as a large integer, consider $k \approx 3/\varepsilon$. We have $(k-1)/(k+2) = 1 - \frac{3}{k+2} \approx 1 - \varepsilon$. Thus, the gap is $\approx 2^{(\log N)^{(1-\varepsilon)}}$.

3 Equivalence between Label Cover variants

3.1 Label Cover variant 1 (Problem A) [ABSS97]

The input is a bipartite graph $G = (V_1, V_2, E \subseteq V_1 \times V_2)$, a set of possible labels B_1 and B_2 for vertices in V_1 and V_2 respectively, and a relation $\Pi \subseteq E \times B_1 \times B_2$ that consists of admissible pairs of labels for each edge. We call the vertices in V_1 *left* vertices, and the vertices in V_2 *right* vertices.

A *labelling* of the graph is a pair of functions $(\mathcal{P}_1, \mathcal{P}_2)$ where $\mathcal{P}_1 : V_1 \rightarrow 2^{B_1}$ and $\mathcal{P}_2 : V_2 \rightarrow 2^{B_2}$. The functions $(\mathcal{P}_1, \mathcal{P}_2)$ assign a set of labels - possibly empty to each vertex in the graph.

The l_1 - *cost* of the labelling is $\sum_{v \in V_1} |\mathcal{P}_1(v)|$ and its l_∞ - *cost* is $\max_{v \in V_1} |\mathcal{P}_1(v)|$.

A labelling is said to *cover* an edge $e = (v_1, v_2)$ if both $\mathcal{P}_1(v_1)$ and $\mathcal{P}_2(v_2)$ are non-empty and *for every label $b_2 \in \mathcal{P}_2(v_2)$, there exists a $b_1 \in \mathcal{P}_1(v_1)$ such that $(e, b_1, b_2) \in \Pi$* .

A *total-cover* of G is a labelling that covers every edge. (*Total cover is a labelling, not a property*)

The decision version of the problem asks to distinguish between YES and NO instances where an instance is a YES instance if there exists a *total-cover* with l_1 - *cost* = $|V_1|$ and l_∞ - *cost* = 1. The instance is a NO instance otherwise.

Remark 31. Note that we do not care much about the labels of right vertices other than the edge constraints. The l_1 - *cost* and l_∞ - *cost* are both defined in terms of left vertices alone.

Remark 32. If the graph contains some vertices which have no edges, then the labels to those vertices do not affect any edge constraints.

Thus, hereafter we assume the degree of each vertex is atleast 1.

Remark 33. If a right vertex has multiple edges incident on it, then each label of that vertex must have a left pre-image (informally) corresponding to each edge, to satisfy all edges.

Consider a right vertex w with 2 edges $\{e_1 = (v_1, w), e_2 = (v_2, w)\}$ incident on it. If the labelling assigns labels $\{\lambda_1, \lambda_2\}$ to w , then there must exist (possibly non-distinct) $\mu_1, \mu_2 \in \mathcal{P}_1(v_1)$ and (possibly non-distinct) $\psi_1, \psi_2 \in \mathcal{P}_1(v_2)$ such that $(e_1, \mu_1, \lambda_1), (e_1, \mu_2, \lambda_2), (e_2, \psi_1, \lambda_1), (e_2, \psi_2, \lambda_2)$ all belong to relation Π , for the labelling to cover both edges e_1, e_2 .

Remark 34. Since we are assuming each vertex has degree atleast 1, the l_1 - *cost* = $|V_1|$ and l_∞ - *cost* = 1 constraints are equivalent.

($\Rightarrow$) l_1 - *cost* = $|V_1|$. Each vertex has degree ≥ 1 , i.e., there is an edge incident on each vertex. This implies label-set for each left vertex is non-empty. Thus, each left vertex has atleast one label. Now, there are $|V_1|$ many left vertices each with at least 1 label such that $\sum_{v \in V_1} |\mathcal{P}_1(v)| = |V_1|$. Therefore each left vertex must have exactly 1 label. This implies l_∞ - *cost* = 1.

($\Leftarrow$) l_∞ - *cost* = 1. Each vertex has degree ≥ 1 and consequently atleast 1 label. Since l_∞ - *cost* = $\max_{v \in V_1} |\mathcal{P}_1(v)| = 1$, each left vertex must have exactly 1 label. This implies l_1 - *cost* = $|V_1|$. Therefore, hereafter we ignore l_∞ - *cost* = 1 constraint.

3.2 Label Cover variant 2 (Problem B)[Kho09]

$G = (V_1, V_2, E \subseteq V_1 \times V_2)$ is a bipartite graph. V_1 : left vertices, V_2 : right vertices, E : edges.

We have to assign exactly one **label** to every vertex. Vertices in V_1 receive label from set B_1 and vertices from V_2 receive label from set B_2 . A *labelling* of the graph is a pair of functions $(\mathcal{P}_1, \mathcal{P}_2)$ where $\mathcal{P}_1 : V_1 \rightarrow B_1$ and $\mathcal{P}_2 : V_2 \rightarrow B_2$. The functions $(\mathcal{P}_1, \mathcal{P}_2)$ assign exactly one label to each vertex in the graph.

Further, there is a relation $\Pi \subseteq E \times B_1 \times B_2$ which consists of admissible pairs of *labels* for each edge.

Labelling $(\mathcal{P}_1, \mathcal{P}_2)$ **satisfies** edge $e = (v, w)$ if:

$$(e, \mathcal{P}_1(v), \mathcal{P}_2(w)) \in \Pi$$

The optimum $OPT(LC)$ of the instance is defined to be the maximum fraction of edges satisfied by any labelling.

The decision version of the problem asks to distinguish between YES and NO instances where an instance is a YES instance if there exists a *labelling* $(\mathcal{P}_1, \mathcal{P}_2)$ that satisfies all edges (i.e., $OPT(LC) = 1$). The instance is a NO instance otherwise.

Remark 35. If the graph contains some vertices which have no edges, then the labels to those vertices do not affect any edge constraints.

Thus, hereafter we assume the degree of each vertex is atleast 1.

3.3 Problem A to Problem B reduction

We provide a constant-time conversion of an arbitrary instance of Problem A to an instance of Problem B such that the YES/NO instances are preserved.

3.3.1 Reduction

Given $G = (V_1, V_2, E), B_1, B_2, \Pi$ for Problem A instance. We construct $G' = (V'_1, V'_2, E'), B'_1, B'_2, \Pi'$ corresponding to Problem B.

For Problem B instance $G' = G, \Pi' = \Pi, B'_1 = B_1, B'_2 = B_2$. All the inputs remain same. We only add “*exactly one label per vertex*” constraint. The problem statement now becomes “*Does there exist a labelling that satisfies all edges?*”

3.3.2 YES instances

If we have a YES instance of Problem A, then there exists a labelling that satisfies all edges and has $l_1 - cost = |V_1|$. Let $(\mathcal{P}_1, \mathcal{P}_2)$ be such a labelling. The labelling assigns exactly one label to each left vertex. Each right vertex has ≥ 1 labels.

Construct a new labelling $(\mathcal{P}'_1, \mathcal{P}'_2)$ as follows:

- $\mathcal{P}'_1 = \mathcal{P}_1$
- For any right vertex w , $\mathcal{P}'_2(w)$ is any one of the right labels amongst $\mathcal{P}_2(w)$. In essence, $\mathcal{P}'_2$ just assigns any one label amongst (possibly multiple) labels assigned by $\mathcal{P}_2$.

$(\mathcal{P}'_1, \mathcal{P}'_2)$ thus constructed is a labelling for corresponding Problem B instance. We will now prove that $(\mathcal{P}'_1, \mathcal{P}'_2)$ covers all edges.

Case 1: We know that $(\mathcal{P}_1, \mathcal{P}_2)$ covers all edges. If for right vertex w , $|\mathcal{P}_2(w)| = 1$, then $\mathcal{P}'_2(w) = \mathcal{P}_2(w)$. All the edges incident on w are satisfied by $(\mathcal{P}'_1, \mathcal{P}'_2)$, since $(\mathcal{P}'_1, \mathcal{P}'_2)$ is same as $(\mathcal{P}_1, \mathcal{P}_2)$ for such edges.

Case 2: Now consider any arbitrary right vertex w' such that $|\mathcal{P}_2(w)| > 1$. Let $\lambda = \mathcal{P}'_2(w)$. We have $\lambda \in \mathcal{P}_2(w)$. Let $e = (v, w)$ be any edge incident on w and let $\mu_0 = \mathcal{P}'_1(v) = \mathcal{P}_1(v)$. By definition of edge satisfaction corresponding to Problem A, we have $(e, \mu_0, \lambda) \in \Pi$. (For right label λ corresponding to w there must exist a left label μ corresponding to v such that $(e, \mu, \lambda) \in \Pi$, but v has only one left label μ_0 . Therefore $(e, \mu_0, \lambda) \in \Pi$)

Thus $(e, \mathcal{P}'_1(v), \mathcal{P}'_2(w)) \in \Pi'$. Therefore all edges incident on w are satisfied by $(\mathcal{P}'_1, \mathcal{P}'_2)$.

Since the above 2 cases are exhaustive, $(\mathcal{P}'_1, \mathcal{P}'_2)$ is a labelling to Problem B instance that satisfies all edges. Hence, the constructed instance is a YES instance.

3.3.3 NO instances

There does not exist a *total-cover* with $l_1 - cost = |V_1|$.

We prove by contradiction. Suppose the constructed Problem B instance is a YES instance. Let

$(\mathcal{P}'_1, \mathcal{P}'_2)$ be the corresponding labelling that covers all edges. Clearly, $(\mathcal{P}'_1, \mathcal{P}'_2)$ is also a valid labelling for original Problem A instance that covers all edges. Further this assignment gives $l_1 - cost = |V_1|$. This implies the original Problem instance is a **YES** instance, a contradiction. Therefore the constructed Problem B instance is a **NO** instance.

This completes the proof of reduction from Problem A to Problem B. There is constant time required for the reduction since the inputs remain same. Further YES/NO instances are preserved.

3.4 Problem B to Problem A reduction

We provide a constant-time conversion of an arbitrary instance of Problem B to an instance of Problem A such that the YES/NO instances are preserved.

3.4.1 Reduction

Given $G = (V_1, V_2, E), B_1, B_2, \Pi$ for Problem B instance. We construct $G' = (V'_1, V'_2, E'), B'_1, B'_2, \Pi'$ corresponding to Problem A.

For Problem A instance $G' = G, \Pi' = \Pi, B'_1 = B_1, B'_2 = B_2$. All the inputs remain same. We only remove the unique label constraint. The problem statement now becomes “*Does there exist a total-cover with $l_1 - cost = |V'_1| = |V_1|$?*”

3.4.2 YES instances

Let $(\mathcal{P}_1, \mathcal{P}_2)$ be a labelling for original Problem B instance that covers all edges. Clearly, $(\mathcal{P}_1, \mathcal{P}_2)$ is also a valid labelling for constructed Problem A instance that covers all edges. Further this assignment gives $l_1 - cost = |V_1|$. This implies the constructed Problem A instance is a YES instance.

3.4.3 NO instances

We prove this by contradiction. Suppose the original Problem B instance is a NO instance, but the constructed Problem A instance is a YES instance.

For constructed Problem A instance, there exists a labelling that satisfies all edges and has $l_1 - cost = |V_1|$. Let $(\mathcal{P}'_1, \mathcal{P}'_2)$ be such a labelling. The labelling assigns exactly one label to each left vertex. Each right vertex has ≥ 1 labels. The same proof as in YES case of Problem A to Problem B reduction gives Problem B instance as a YES instance, a contradiction. Therefore, the constructed Problem A instance is a NO instance.

This completes the proof of reduction from Problem B to Problem A. There is constant time required for the reduction. Further YES/NO instances are preserved.

Hence, Problem A and Problem B are equivalent, or, the 2 versions of Label Cover Problem are equivalent.

4 Minimization to Maximization variant

Consider the multiple-labels variant of the label cover problem (Problem A). Given $G = (V_1, V_2, E)$, B_1, B_2, Π for Problem A instance. Let $|V_1| = n$, $|V_2| = m$. We further have

- Every left-vertex (V_1 vertices) has same degree k . ($\Rightarrow |E| = k \cdot |V_1|$)
- $|V_1| \geq |V_2|$ (defining so because the unique-label variant has this property)
- Further, for each $e \in E$ and $b_1 \in B_1$, there is atmost one $b_2 \in B_2$ such that $(e, b_1, b_2) \in \Pi$ (Function criterion)
- Each right vertex has exactly one label. (Proven earlier that assigning multiple labels to right vertices doesn't cover any extra edges.)

Consider a total-cover $(\mathcal{P}_1, \mathcal{P}_2)$ with $l_1 - cost = \gamma \cdot |V_1|$. Let $v_1, v_2, \dots, v_n$ be the left vertices. Let S_{v_i} denote the set of labels assigned to left-vertex v_i , $i = 1, 2, \dots, n$.

We have $\sum_{i=1}^n |S_{v_i}| = \gamma \cdot n$

Consider an instance of unique-label variant of the label cover problem (Problem B) with same $G = (V_1, V_2, E), B_1, B_2, \Pi$ as Problem A instance. We want to see how a solution of Problem A instance transforms to a solution of Problem B. In particular we want to see that how does " $l_1 - cost = \gamma \cdot n$ " transforms to fraction λ of edges satisfied in unique-label variant.

Construct a labelling for Problem B instance as follows:

- Assign each right vertex in Problem B instance the same label as in Problem A instance.
- For each left-vertex v_i , the set of labels S_{v_i} satisfies k edges. This implies, by Pigeonhole Principle, there exists a label $b_{v_i} \in S_{v_i}$ that satisfies atleast $k/|S_{v_i}|$ edges. Assign that label b_{v_i} to the left-vertex v_i in Problem B instance.

Let us analyze how many edges are satisfied in Problem B labelling described above.

Total number of edges satisfied $\geq \sum_{i=1}^n k/|S_{v_i}|$

By $AM - HM$ inequality, we have:

$$\begin{aligned} \frac{\sum_{i=1}^n |S_{v_i}|}{n} &\geq \frac{n}{\sum_{i=1}^n 1/|S_{v_i}|} \\ \Rightarrow \gamma &\geq \frac{n}{\sum_{i=1}^n 1/|S_{v_i}|} \\ &\Rightarrow \sum_{i=1}^n \frac{1}{|S_{v_i}|} \geq \frac{n}{\gamma} \end{aligned} \tag{10}$$

$$\text{Total number of edges satisfied} \geq \sum_{i=1}^n k/|S_{v_i}| \geq \frac{k \cdot n}{\gamma}$$

This implies atleast $1/\gamma$ fraction of edges can be satisfied. ($\lambda \geq 1/\gamma$)

4.1 Tightness of bound

We prove that if $\gamma \cdot |V_1|$ is the minimum $l_1 - cost$ for any labelling in Problem A instance, then for corresponding Problem B instance we can only guarantee that there exists a labelling that satisfies atleast $1/\gamma$ edges. We cannot improve this lower-bound. In particular, we give an example where this lower bound is attained.

Note that we are asking $\gamma \cdot |V_1|$ to be the least $l_1 - cost$ amongst all labellings for the following reason. If for Problem A instance, a non-optimal labelling gives $l_1 - cost = \gamma_1 \cdot n$ and the optimal labelling gives $l_1 - cost = \gamma_2 \cdot n$ with $\gamma_1 > \gamma_2$, then our result corresponding to non-optimal labelling says atleast $1/\gamma_1$ fraction of edges are satisfied in corresponding Problem B instance, and for optimal labelling we get atleast $1/\gamma_2$ fraction of edges is satisfied in the corresponding Problem B instance. Here, we see that the $1/\gamma_1$ lower bound was not tight. There existed a tighter $1/\gamma_2$ lower bound. Thus, the tightness of lower bound can be established only when considering optimal labelling in Problem A instance.

4.1.1 Example

- $V_1 = \{a, b\}$ (left vertices)
- $V_2 = \{x, y\}$ (right vertices)
- $E = \{e_1 = (a, x), e_2 = (a, y), e_3 = (b, x), e_4 = (b, y)\}$
- $B_1 = \{odd, even\}$
- $B_2 = \{0, 1\}$
- $\Pi = \{(e_1, odd, 1), (e_2, even, 0), (e_3, odd, 1), (e_4, odd, 0)\}$

Assigning $a : (odd, even)$; $b : (odd, even)$; $x : (1)$; $y : (0)$ satisfies all edges and this is optimal. Here $\gamma = 2$. For the corresponding unique-label (Problem B) instance, we can satisfy atmost 2 edges for any labelling. That is $1/\gamma$ fraction. Thus, the $1/\gamma$ lower bound is tight.

5 Maximization to Minimization variant

Consider the unique-label variant of the label cover problem (Problem B). Given $G = (V_1, V_2, E)$, B_1, B_2, Π for Problem B instance. Let $|V_1| = n$, $|V_2| = m$. We further have

- Every left-vertex (V_1 vertices) has same degree k . ($\Rightarrow |E| = k \cdot |V_1|$)
- $|V_1| \geq |V_2|$
- Further, for each $e \in E$ and $b_1 \in B_1$, there is atmost one $b_2 \in B_2$ such that $(e, b_1, b_2) \in \Pi$ (Function criterion)

Let $v_1, v_2, \dots, v_n$ be the left vertices. Let $(\mathcal{P}_1, \mathcal{P}_2)$ be a labelling that satisfies λ fraction of edges. Consider an instance of multiple-labels variant of the label cover problem (Problem A) with same $G = (V_1, V_2, E)$, B_1, B_2, Π as Problem B instance. We want to see how a solution of Problem B instance transforms to a solution of Problem A. In particular we want to see that how does “ λ fraction of edges satisfied” transforms to $l_1 - cost$ in multiple-label variant.

$(\mathcal{P}_1, \mathcal{P}_2)$ satisfies $\lambda \cdot |E| = \lambda \cdot k \cdot |V_1|$ edges. This implies $(1 - \lambda) \cdot k \cdot |V_1|$ edges are unsatisfied. We construct a total-cover for Problem A (multiple-labels) instance using $(\mathcal{P}_1, \mathcal{P}_2)$.

$|V_1|$ left-labels (as per $(\mathcal{P}_1, \mathcal{P}_2)$) satisfy $\lambda \cdot k \cdot |V_1|$ edges. For satisfying remaining $(1 - \lambda) \cdot k \cdot |V_1|$ edges, we need atmost 1 additional label per edge. Thus, in atmost $|V_1| + (1 - \lambda) \cdot k \cdot |V_1|$ left-labels, we get a total-cover. This implies $l_1 - cost \leq |V_1|(1 + (1 - \lambda) \cdot k)$.

This implies $\gamma \leq 1 + (1 - \lambda) \cdot k$

5.1 Tightness of bound

Once again, the bound is tight only if λ fraction is corresponding to optimal solution of Problem B. We prove that if λ is the maximum fraction of edges satisfied by any labelling for Problem B instance, then for corresponding Problem A instance we can only guarantee that $\gamma \leq 1 + (1-\lambda) \cdot k$. We cannot improve this upper-bound on γ . In particular, we give an example where this upper bound is attained.

5.1.1 Example

- $V_1 = \{a, b\}$ (left vertices)
- $V_2 = \{x, y\}$ (right vertices)
- $E = \{e_1 = (a, x), e_2 = (a, y), e_3 = (b, x), e_4 = (b, y)\}$
- $B_1 = \{\text{odd}, \text{even}\}$
- $B_2 = \{0, 1\}$
- $\Pi = \{(e_1, \text{odd}, 1), (e_2, \text{even}, 0), (e_3, \text{odd}, 1), (e_4, \text{odd}, 0)\}$

Any unique-label assignment satisfies atmost 2 edges. That is $\lambda = 1/2$.

Assigning $a : (\text{odd}, \text{even})$; $b : (\text{odd}, \text{even})$; $x : (1)$; $y : (0)$ satisfies all edges and this is optimal. $\gamma = 2 = 1 + (1 - \lambda) \cdot k$. Thus the bound $\gamma \leq 1 + (1 - \lambda) \cdot k$ is tight.

5.2 Observations

5.2.1 Obs. 1

We have established:

- $\gamma \leq 1 + (1 - \lambda) \cdot k$ for optimal λ (multiple-labels to unique-label)
- $\lambda \geq 1/\gamma$ for optimal γ . (unique-label to multiple-labels)

Here, the comparators are $\leq$ and $\geq$ respectively. There exist examples that show these comparators are not $=$ even for optimal γ, λ . (The examples given above might give an impression that we get equality when considering optimal values of γ, λ . But, this is not always the case.)

5.2.2 Obs. 2

The 2 variants are inherently different. Optimizing one problem does not optimize the other problem.

For example, consider an instance where left vertices have degree 200 each.

Let $(\mathcal{P}_1, \mathcal{P}_2)$ be a total-cover that assigns 4 labels to each left vertex, and each label satisfies 50 distinct edges.

Let $(\mathcal{Q}_1, \mathcal{Q}_2)$ be a total-cover that assigns 101 labels to each left vertex. Amongst the 101 labels, one label satisfies 100 edges and rest all 100 labels satisfy 1 edge each.

Now, $l_1 - \text{cost}$ corresponding to $(\mathcal{P}_1, \mathcal{P}_2)$ is $4 \cdot |V_1|$ whereas $l_1 - \text{cost}$ for $(\mathcal{Q}_1, \mathcal{Q}_2)$ is $101 \cdot |V_1|$. Clearly,

$(\mathcal{P}_1, \mathcal{P}_2)$ is better than $(\mathcal{Q}_1, \mathcal{Q}_2)$ with respect to $l_1 - cost - minimization$ variant (Problem A). But, $(\mathcal{Q}_1, \mathcal{Q}_2)$ is better than $(\mathcal{P}_1, \mathcal{P}_2)$ with respect to *unique labels (maximization)* variant (Problem B).

6 Label Cover Hardness Results

- We have a reduction from **LabelCover(maximization)** to CVP. The initial Label Cover instance satisfies *function* condition and the graph is left-regular [Kho09]. Here, the gap $R^{-\beta}$ of Label Cover translates to gap $g = \frac{1}{20}R^{\beta/2}$ for CVP. R is the size of left vertex set, that is assumed to be ≥ 7 , and $\beta > 0$ is an absolute constant.
- We have a reduction from **LabelCover(minimization)** to CVP. The initial Label Cover instance satisfies *function* condition and the graph is left-regular [ABSS97]. The optimum in the CVP instance has cost $\geq \text{minimum } l_1 - \text{cost}$ of total-cover. Further, if there is a total-cover with $l_1 - \text{cost} = |V_1|$, the the optimum is also $|V_1|$.
- If CVP can be solved in time $f(n)$, then LC can be solved in time $\text{poly}(n) \cdot O(f(\text{poly}(n)))$.
- If LC cannot be solved in time $g(n)$, then it implies CVP cannot be solved in time $g(n)$ either (*not exactly $g(n)$, but referring to complexity classes in general, like polynomial time, quasi-polynomial time, et cetera.*).
- Thus “*Hardness of LC translates to hardness of CVP*” (applicable only when considering LC with function condition, and left-regular graph.)
- “Cannot do” - type results for LC translate to “Cannot do” - type results for CVP. These are the results we are after.
- “Cannot do” type results for general LC (without function condition) are not very important for us, because we cannot directly translate these results to get hardness for CVP. (*Maybe the hardness/cannot-do is for only non-function-condition LC instances.*)
- If we have a “can-do” type result for general LC i.e., if we can solve a general LC instance in time T , then we can definitely solve LC instance with function condition, left-regular graph in time T as well. Nevertheless, these results do not give any bounds on hardness of CVP.
- Thus, if we have “Cannot” result, we want it for LC with function condition, left-regular graph. For “Can” result, we can work with general LC as well.

6.1 Results

- **Arora et. al ’1997** [ABSS97]:
 - For each $\delta > 0$, there exists quasi-polynomial time reduction from SAT to **LabelCover(minimization)** (decision version) instance of size N with Gap $g = 2^{(\log N)^{0.5-\delta}}$.
 - The YES instances have a total-cover with $l_1 - \text{cost} = |V_1|$, and for NO instances any total-cover has $l_1 - \text{cost} \geq g \cdot |V_1|$.
 - By quasi-polynomial we mean $2^{\text{poly}(\log n)}$. We have this result for function-condition abiding, left-regular graph variant of LC.
 - This implies *LC* with aforementioned Gap g cannot be solved in *quasi-polynomial time* unless $NP \subseteq DTIME(2^{\text{poly}(\log n)})$.

- $(\text{Gap } g) - LC$ is quasi-polynomial hard means if we have a quasi-polynomial time algorithm to solve $(\text{Gap } g) - LC$, then SAT can also be solved in quasi-polynomial time.
- Quasi-polynomial time reduction from SAT to LC with gap g means that it is quasi-polynomial hard to distinguish between the two cases.
- This gap directly translates to an approximation factor of ' g '. If you had an algorithm that could approximate the minimum number of labels within a factor better than ' g ', you could distinguish between the YES and NO cases.
- Since distinguishing between the YES and NO cases is quasi-polynomial hard (given the reduction from SAT), approximating the minimum number of labels within a factor better than g is also quasi-polynomial hard.
- It means that it's quasi-polynomial hard to find a solution that uses fewer than g times the optimal number of labels.
- In other words, it is quasi-polynomial hard to approximate the minimum number of left labels within a factor of g .

• **Dinur, Safra '2004**[\[DS04\]](#):

- General **LabelCover**(minimization)(without function condition) is NP-hard to approximate within $\gamma = 2^{(\log n)^{1-\delta}}$ where $\delta = (\log \log n)^{-c}$, for any $0 < c < 1/2$.
- This result achieves NP-hardness. That is no *polynomial-time algorithm* can give a solution that is atmost γ times the optimal $l_1 - cost$ unless $P=NP$.
- Compared to previous result, this result obtains NP-hardness (which is an improvement on quasi-polynomial hardness)
- The approximation factor γ is also an improvement upon the previous factor g .

7 Comments on Parallel Reduction

- *One-round two-player game G :*
Given bipartite graph $G = (U, V, E \subseteq U \times V)$, and constraints $\pi \subseteq \Sigma \times \Sigma$ for an alphabet Σ . The **value** of a game is the maximum over all assignments $f : U \rightarrow \Sigma, g : V \rightarrow \Sigma$ of the fraction of edges (constraints) satisfied. A constraint π is satisfied if $(f(u), g(v)) \in \pi_{u,v}$. Same setting as maximization version of Label Cover. No function condition on projection constraints.
- Can visualise this as a game G where a referee interacts with 2 players, Alice and Bob. Alice has some strategy $f : U \rightarrow \Sigma$, and Bob has some strategy $g : V \rightarrow \Sigma$. Alice and Bob cannot communicate during the game (can collectively decide strategies beforehand). Referee randomly selects an edge $u, v \in E$ and sends u to Alice and v to Bob. Alice responds with $f(u)$ and Bob responds with $g(v)$. They succeed if $(f(u), g(v)) \in \pi_{u,v}$.
- *k -fold repetition* of the game G involves referee choosing k edges independently from E , and playing the game accordingly. Alice and Bob succeed if all the k edges are satisfied.
- In the k -fold repetition game, if the strategy for any vertex does not depend on the other vertices received, then $value(G^k) = value(G)^k$ (If Alice and Bob are looking at each vertex in isolation when deciding its label, then the best they can do is $value(G)^k$). But the strategy for any vertex may employ information gained from seeing other vertices as well. We seek for bound on $value(G^k)$ when such strategies are allowed.
- Previous results:
 - Parallel Repetition theorem. [Raz95]
 $value(G) \leq 1 - \varepsilon \Rightarrow value(G^k) \leq (1 - \varepsilon^{32}/2)^k$. The value decays exponentially with the number of repetitions k . [$0 < \varepsilon < 1 \Rightarrow \varepsilon^{32}/2 \ll \varepsilon$. Thus, $(1 - \varepsilon^{32}/2)^k$ is a very relaxed bound. We would like to bring this bound as close to $(1 - \varepsilon)^k$ as possible.]
 - current best. [Rao08]
 $value(G) \leq 1 - \varepsilon \Rightarrow value(G^k) \leq (1 - \varepsilon^2/2)^k$
- Unaddressed gaps:
 - What happens if $value(G)$ is subconstant, i.e., $value(G) = o(1)$? Previous expression doesn't give good usable bounds.
 - What if we have very low number of repetitions? [$k \ll 1/\varepsilon^2$]?
- This paper's contributions: [DS14]
 - *low value*: $value(G) \leq \rho \Rightarrow value(G^k) \leq (2\rho)^{k/2}$
 - *few repetitions* [$k \ll 1/\varepsilon^2$]: $value(G) \leq 1 - \varepsilon \Rightarrow value(G^k) \leq (1 - \varepsilon)^{\sqrt{k}}$
- Authors use a New Parameter called *relaxed value* (denoted by $relax(.)$) instead of *value*. Value is not very handy. We need certain guarantees about *relax*:
 - $relax(G) \geq value(G)$ for all G

- $relax(G^k) = relax(G)^k$ for all G
- $relax(G)$ is a good approximation of $value(G)$ for all G
- Analytical setup. Convert the graph to *Label-extended graph*. Expand each vertex to $|\Sigma|$ vertices, one corresponding to each label from Σ . Thus we have $|U| \times |\Sigma|$ left vertices and $|V| \times |\Sigma|$ right vertices. For each original edge (u, v) do the following. $((u, \alpha), (v, \beta))$ is an edge iff $(\alpha, \beta) \in \pi_{u,v}$, i.e., (α, β) labels satisfy the edge (u, v) in original graph.
- Let $f : U \times \Sigma \rightarrow \mathbb{R}$ be a function on the left vertex set of label extended graph. We can define a linear operator $G : \mathbb{R}^{U \times \Sigma} \rightarrow \mathbb{R}^{V \times \Sigma}$ corresponding to the label extended graph such that:

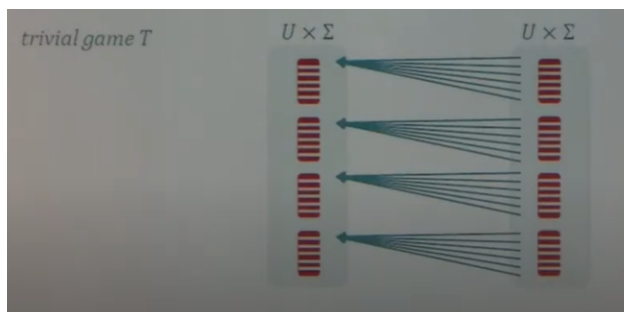
$$Gf(v, \beta) = \mathbb{E}_{u: (u,v) \in E} \left(\sum_{\alpha: (\alpha, \beta) \in \pi_{u,v}} f(u, \alpha) \right)$$

G is basically the adjacency matrix of the label-extended graph. The function $Gf(v, \beta)$ averages over all the left-neighbours u of v , the following quantity: Sum of $f(u, \alpha)$ for all α such that (α, β) satisfy the edge (u, v) .

Given right vertex v , right label β , do the following:

- First find all the left neighbours u of v .
- For each such u , find all α such that $(\alpha, \beta) \in \pi_{u,v}$ [(α, β) satisfy the edge (u, v)]
- Then for each u , compute the summation over all α , of $f(u, \alpha)$. We get a value corresponding to each u .
- Now compute the average over all u , of the value found in previous step.
- This is the value $Gf(v, \beta)$.
- $f : U \times \Sigma \rightarrow \mathbb{R}$ is called an assignment if $f \geq 0$, and $\sum_{\alpha} f(u, \alpha) = 1$ for all $u \in U$. Here we are allowing fractional values as well.
- For assignment f , $Gf(v, \beta)$ is the probability that random “v-neighbour” demands symbol β to satisfy its edge.
- Lots of analysis.
- Tensor product of Operators defined above is equivalent to parallel repetition.
- $\|Gf\|_2^2 = \mathbb{E}_v \sum_{\beta} Gf(v, \beta)^2$
- Good Proxy for $value(G)$ is $\max_{assignment f} \|Gf\|_2$ (Recall $relax(.)$ described before)
- $value(G) \leq \max_{assignment f} \|Gf\|_2 \leq value(G)^{1/2}$ (A claim the authors prove)
- Prove some theorems using lots of analysis. Two interesting techniques used:

- Defining a trivial game T . Same left and right vertex sets. Only the same vertices on the left side and right side have an edge between them. The label-extended graph is such that exactly 1 particular labelling of left vertices satisfies all edges irrespective of right labellings. Thus, in this trivial game if Alice follows her strategy, then irrespective of Bob's answers, they succeed the game with probability 1.



- Using hybrid arguments.

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